

CHANDRAPAL

Bengaluru, Karnataka — +91-7905866596 — 2023chandra.pal@vidyashilp.edu.in
[LinkedIn](#) — [GitHub](#)

PROFESSIONAL SUMMARY

AI/ML Engineer and Data Science undergraduate focused on building intelligent systems across **deep learning, Generative AI, agentic workflows, and AI systems engineering**. Hands-on experience developing specification-driven cognitive AI platforms, on-device lifelong memory systems, **LLM-powered RAG applications, semantic retrieval pipelines, and multimodal AI solutions**, with additional experience in financial analytics, optimization, and applied machine learning.

EDUCATION

Vidyashilp University, Bengaluru

2023–2027

B.Tech. in Data Science (Hons.) + Minor in Finance

CGPA: 7.89/10

PROJECTS

The Inevitable – Specification-Driven Cognitive AI Platform

TypeScript — npm — Turborepo — JSON Schema —

Event-Driven Architecture — Knowledge Graphs — Agent Orchestration — OpenTelemetry

- Architected a **specification-driven Cognitive AI platform** for personalized learning, knowledge construction, mastery, research, and intellectual development using persistent cognitive state and specialized AI agents.
- Developed architecture for **agent orchestration, prerequisite-aware learning, adaptive explanation, cognitive memory, knowledge-graph reasoning, multimodal cognition, and replayable event-driven state**.

EdgeMemory – On-Device Lifelong Memory AI

Python — PyTorch — ASR — Embeddings — Vector Retrieval — Deep

Learning — ONNX — Model Optimization

- Built a privacy-first **lifelong memory pipeline** converting voice reflections into timestamped semantic memories using ASR, embeddings, and time-aware retrieval.
- Optimized models using **knowledge distillation, pruning, quantization, and ONNX inference** for constrained environments.

ResuWhisperAI – Generative AI Resume Intelligence System

Python — LLMs — RAG — Embeddings — Semantic

Search — Speech/ASR — Structured Generation

- Developed a **Generative AI resume intelligence system** combining resume parsing, LLM reasoning, semantic retrieval, and structured candidate profiling.
- Implemented **RAG pipelines** using document processing, embeddings, semantic retrieval, and contextual LLM generation.

Healthcare AI Assistant – Multimodal Clinical Decision-Support Prototype

Python — Machine Learning — Google

Gemini — Voice AI — Probability Ranking — Interactive Dashboards

- Developed a multimodal healthcare prototype accepting **text, voice, and body-map inputs** for ranked predictions across **41 diseases and 132 symptoms**.
- Integrated **symptom weighting and probability ranking** with Google Gemini for contextual explanations and recommendations.
- Built an interactive workflow combining **voice interaction, ML inference, conversational AI, and dashboards**.

EXPERIENCE

Data Science & Analytics Intern

Apr 2025 – Jun 2025

Zio Development

- Developed an end-to-end **AI-assisted disease prediction pipeline** covering **41 diseases and 132 symptoms**, combining ML probability ranking with Google Gemini explanations and recommendations.
- Implemented **symptom weighting, probabilistic ranking, and inference workflows** for ranked disease predictions.
- Built an interactive AI application integrating **voice input, ML inference, conversational AI, and real-time dashboards**.

RESEARCH & OPTIMIZATION

Carbon Footprint Optimization – Diet Optimization Research

Python — Linear Programming — Optimization —

Sustainability Analytics

- Formulated a **linear programming model** balancing nutrition, food cost, and carbon emissions for personalized low-emission meal plans.
- Developed a data-driven prototype translating optimization results into **cost- and carbon-efficient dietary recommendations**.

TECHNICAL SKILLS

Programming: Python, TypeScript, Java, C, SQL **ML/DL:** PyTorch, scikit-learn, NumPy, Pandas, CNNs, RNNs, GRUs, LSTMs, Attention Mechanisms, Transformers, Model Evaluation, Optimization

Generative AI: LLMs, RAG, Embeddings, Semantic Search, Prompt Engineering, Structured Generation, AI Agents, Agentic Workflows

AI Systems: Knowledge Graphs, Agent Orchestration, Event-Driven Architecture, Cognitive Memory, State Management, AI Observability, OpenTelemetry

Backend & Data: FastAPI, REST APIs, PostgreSQL, MySQL, Vector Search, Streamlit, Plotly

MLOps & Deployment: Docker, Git, CI/CD, MLflow, ONNX, Model Quantization, Model Pruning

ACHIEVEMENTS

- Top 7 Winning Teams**, The AI Hiring Show – Vibe Coding, Power Hiring, 2025
- Shiv Nadar Foundation (HCL) Scholar** – Academic Merit & Leadership